

Proficiency Test 2

Task 1

```
for (i in 1:3) {  
  print(mtcars$mpg[i])  
}
```

```
[1] 21  
[1] 21  
[1] 22.8
```

Task 2

```
count_above <- function(vector, cutoff) { # see explanation L24  
  count <- 0 # creates a "count" for the scenario below  
  
  for (i in 1:length(vector)) { # takes that for every vector,  
    if (vector > cutoff) { # if a vector is greater than cutoff  
      count <- count + 1 # then add to the count  
    }  
  }  
  return(count) # resets the count to 0  
}  
# creates a function named count_above containing the variables  
# vector and cutoff.
```

```
for (i in 1:nrow(iris)) {  
  if (iris$Sepal.Length[i] > 6) {  
    print(iris$Sepal.Length)  }  
}
```

[illegible]

[illegible]

```
# The code printed no Sepal Lengths, so there are no iris flowers
# in the dataset with sepal lengths greater than 6 cm.
```

```
for (i in 1:nrow(mtcars)) {
  if (mtcars$disp[i] >= 150) {
    print(mtcars$disp)
  }
}
```

[1]	160.0	160.0	108.0	258.0	360.0	225.0	360.0	146.7	140.8	167.6	167.6	275.8
[13]	275.8	275.8	472.0	460.0	440.0	78.7	75.7	71.1	120.1	318.0	304.0	350.0
[25]	400.0	79.0	120.3	95.1	351.0	145.0	301.0	121.0				
[1]	160.0	160.0	108.0	258.0	360.0	225.0	360.0	146.7	140.8	167.6	167.6	275.8
[13]	275.8	275.8	472.0	460.0	440.0	78.7	75.7	71.1	120.1	318.0	304.0	350.0
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[1]	160.0	160.0	108.0	258.0	360.0	225.0	360.0	146.7	140.8	167.6	167.6	275.8

[illegible]

```
[25] 400.0  79.0 120.3  95.1 351.0 145.0 301.0 121.0
[1] 160.0 160.0 108.0 258.0 360.0 225.0 360.0 146.7 140.8 167.6 167.6 275.8
[13] 275.8 275.8 472.0 460.0 440.0  78.7  75.7  71.1 120.1 318.0 304.0 350.0
[25] 400.0  79.0 120.3  95.1 351.0 145.0 301.0 121.0
[1] 160.0 160.0 108.0 258.0 360.0 225.0 360.0 146.7 140.8 167.6 167.6 275.8
[13] 275.8 275.8 472.0 460.0 440.0  78.7  75.7  71.1 120.1 318.0 304.0 350.0
[25] 400.0  79.0 120.3  95.1 351.0 145.0 301.0 121.0
```

```
# The code printed 4 rows of cars that had a disp greater than
# or equal to 150.
```

Task 3

Masking functions involves using functions such as `eval`: `FALSE` or `echo`: `FALSE` to mask other functions' output to make the rendered document “clean” and more readable.

Task 4

```
iris$Sepal.Length[1] # The code returns the sepal length value of # the first row in the data
```

```
[1] 5.1
```

```
iris[1,1]
```

```
[1] 5.1
```